Requirements Alignment and Acceptance Criteria Assessment

Hospital Patient Referral, Treatment and Administration System

Portfolio submission — strategic process modelling and written evaluation

Module

Business Process Modelling and Enterprise Architecture

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1. Purpose and scope

This document accompanies the modelling portfolio. It covers the two pieces of written assessment the brief asks for: a review of how far the models we built meet the needs of the people who use the service, supported by a requirements traceability table, and a review of the acceptance criteria we set, what we changed after testing them, and what we found.

The artefacts reviewed are listed below. All of them are submitted in editable form.

Ref	Artefact	File	What it contains
A1	Strategic BPMN model	UFCEP6-0-3_Hospital_Patient_Pathway_Strategic_R2.bpmn	7 participants, 3 responsibility lanes, 10 business stages, 17 message flows
A2	i* Strategic Dependency model (SD)	SD diagram.txt / SD diagram.svg	15 actors and 28 dependency relationships
A3	i* Strategic Rationale model (SR)	SR diagram.txt / SR diagram.svg	Internal rationale for six actors; 91 links
A4	Case study	Hospital Patient Referral, Treatment and Administration System	The source of the requirements and of the acceptance criteria

One point is worth stating before the detail. The BPMN in this portfolio is a strategic, level-one model. It shows the business stages, which responsibility each one sits under, and the messages that cross the organisational boundary. It does not contain user tasks, service tasks, timers or data objects, because those belong to the operational model that is built separately. That decision affects how some requirements can be evidenced, and the table in section 2.2 is explicit about it: where a rule sits in the stage specification rather than on the diagram, it is recorded as specified, not as modelled.

2. Requirements alignment assessment

2.1 How the trace was carried out

We traced from the i* models rather than from the case study text. The SD and SR already set out whose goals are in play and what the actors depend on each other for, so they gave us a list of intentions to test against the design. For each goal or softgoal we wrote down what the system would have to do, then went looking for that requirement in the strategic BPMN. Three outcomes were possible:

- Modelled — the requirement can be read from the model itself: an element exists for it, or the model's topology carries it (a lane assignment, the ordering of sequence flows, the only path through a gateway).
- Specified — the rule is carried by the documentation attached to the stage that owns it, and the mechanics are implemented one level down in the operational model.
- Assigned — the requirement is not a process-flow concern and belongs to another level or to the platform.

Three checks were run over the model files alongside the trace: a static inspection of the BPMN XML and its diagram interchange, a reachability check over the sequence-flow graph, and a lane-containment check. Section 3.3 reports the results of the same checks in full.

2.2 Requirements traceability table

Ref	Stakeholder and goal	What the system has to do	Where it sits in the strategic BPMN	Evidence	Alignment
R01	Referring GP or hospital — my	The service needs an entry point a referral can arrive through, carrying the referral and its	Start_Referral (message start	The start event is a message event bound to Message_Referral and is the target of MF_ReferralIn from	Modelled

	patient is seen in time	supporting documents	event); MF_ReferralIn	P_Referrer	
R02	Medical Secretaries — a complete pack before review	The receiving team must check the pack against what the hospital needs and be able to ask the referrer for anything missing	T_Check; MF_InformationRequest	T_Check is a call activity bound to medical-secretaries and MF_InformationRequest runs from it to P_Referrer	Modelled
R03	Consultant — the right decision on the referral	Only an authorised consultant acceptance may lead to an appointment, and the decision, the reason and the decision maker must be recorded and returned	T_Review; G_Accepted; MF_ReferralOutcome	G_Accepted is the only gateway on the path into T_Book, Flow_Accepted is the only incoming flow to T_Book and it carries the condition, and MF_ReferralOutcome is named for the decision, the reason and the decision maker	Modelled
R04	All — clinical, administrative and financial responsibility stay apart	Clinical decisions must not be taken by administrative or finance staff, and finance must not decide clinical need	Three responsibility lanes; both gateways in the clinical lane; only T_Funding in the financial lane	Lane membership checked against the laneSet: G_Accepted and G_Treat are in Lane_clinical and Lane_financial holds no decision node	Modelled
R05	Outpatient Bookings — the right appointment, soon	Booking must go through the external scheduling service and the patient must be told once it is confirmed	T_Book; MF_SlotSearch; MF_AppointmentLetter	Both message flows leave T_Book, to P_Scheduling and P_Correspondence	Modelled
R06	Patient — a clear date and time	An appointment inside two weeks also needs a telephone call, and each contact attempt and its outcome has to be recorded	T_Book; MF_ShortNoticeCall	MF_ShortNoticeCall is present as a separate message flow to P_Patient; the two-week trigger and the contact record are set out in the T_Book specification	Specified
R07	Case study ¶6 — urgent referrals and no suitable slot	Referrals that cannot wait for normal administrative timescales, and cases where no slot fits the requested period, must be flagged	T_Book specification	Both situations are stated in the specification. They are not branched on the diagram because the case study never settles the rules for deciding urgency	Specified
R08	Consultant — safe and right treatment	Consent must be recorded and the treatment request authorised before administrative staff may act on it	T_Consult; G_Treat	Flow_5 runs from T_Consult into G_Treat and only Flow_TreatmentAgreed continues into the treatment stages	Modelled
R09	Consultant and Finance — money ready before treatment	A treatment appointment must not be confirmed until payment, an exemption, a funding approval or a payment arrangement is recorded	T_Funding before T_Treatment (Flow_6, Flow_7)	The constraint is a property of the sequence flows: T_Funding → T_Treatment, so it is readable straight off the diagram	Modelled
R10	Finance — knowing who pays	Establish whether the hospital, an approved insurer or the patient pays, send a secure payment request, and record the status and transaction reference	T_Funding; MF_PaymentRequest; MF_PaymentStatus	Two message flows in opposite directions between T_Funding and P_Payment	Modelled
R11	Patient — a refund when treatment is cancelled	Where treatment is cancelled, postponed or changed after payment, finance decides and records any refund; clinical staff must not approve refunds	T_Funding (named for refunds) and its specification	The stage is named 'Determine funding, take payment and resolve refunds'; the refund rule and the separation of duties are in the specification	Specified
R12	Case study ¶11–12 — payment problems	Declined, duplicated or incomplete payments, and payments taken without a confirmation coming back, must be marked for investigation rather than charged again; clinical staff may authorise urgent treatment and must record why	T_Funding specification	The failure handling and the clinical override are both written into the stage specification	Specified
R13	Treatment Bookings — planning the treatment series	Capacity must be obtained from the external treatment, laboratory and imaging services; if a service is unavailable the booking stays pending and no duplicate is created	T_Treatment; MF_CapacityCheck; MF_CapacityConfirm	A request and a reply message flow connect T_Treatment and P_ClinicalServices	Modelled

R14	Case study ¶14–15 — cycle reviews and treatment changes	Each cycle is reviewed clinically before the next is booked, and any change to the plan needs a formal authorisation	T_Treatment specification	Both rules are recorded in the specification, including the referral of financially significant changes to Finance	Specified
R15	Pathway Coordinators — the clinic letter on time	Letters must be completed, approved, processed and sent within seven days, and overdue letters must feed the monitoring and escalation path	T_Letter; T_Monitor specification	T_Letter exists as its own administrative stage; the seven-day, one-month and three-month thresholds are recorded in the T_Monitor specification	Specified
R16	Case study ¶24 — follow-up	A follow-up requested by the clinical team must be arranged inside the period asked for, and a case that does not fit must be highlighted and passed to the pathway team	T_FollowUp; MF_FollowUpSlotSearch	T_FollowUp is a separate stage and its search request runs to P_Scheduling	Modelled
R17	Case study ¶25 — cancellations and missed appointments	Cancellations, declines and non-attendance must be recorded and a decision taken on rebooking, clinical review or telling the referrer; a paid appointment goes to Finance	T_FollowUp (name and specification)	The stage is named for rebooking cancelled or missed appointments; the decision rules are in its specification	Specified
R18	Call Handling — every enquiry answered by the right team	Every enquiry must be recorded, classified, given a priority and routed, and call handlers must not give clinical advice	Start_Enquiry; T_Enquiry; End_Enquiry; specification	The enquiry pathway has its own message start event, stage and end event; the routing rules and the restriction on clinical advice are in the specification	Specified
R19	Management — visibility of the pathway	Management reporting on referral volumes, waiting times, delayed letters, unresolved enquiries, failed payments, refunds and pathway progress	T_Monitor specification	The report list is written into the T_Monitor specification	Specified
R20	All — audit and access control (case study ¶26–27)	Record the user, date, time and nature of each action; audit records must not be editable by ordinary users; users must only reach the information their role needs	Handled at the levels that own these controls	These are platform and non-functional controls rather than process flow, so they are not repeated on a strategic diagram. They are met by the operational process (user task assignment and authorisation), by the Camunda Forms that validate input, and by the engine's own audit trail and identity configuration	Assigned

2.3 What the trace shows

Of the twenty requirements traced, ten are modelled, nine are specified and one is assigned. That split is worth explaining rather than glossing over, because it is a direct consequence of the level we chose to model at.

The ten modelled requirements are the ones a strategic process model is good at. They are structural claims: who is allowed to do what, in which order, and which messages leave the organisation. R04 is the clearest example. The case study spends a lot of words insisting that clinical judgement, administrative work and financial work stay apart, and the model answers that with three lanes, both decision gateways sitting in the clinical lane, and a financial lane that contains one stage and no decisions at all. Nothing in that answer depends on reading the documentation. R09 is similar: the rule that money must be settled before a treatment appointment is confirmed is expressed as the order of two sequence flows, so a reviewer can verify it by looking at the diagram.

The nine specified requirements are real requirements that a level-one model cannot carry on its own: trigger conditions, thresholds, exception handling and the record-keeping that goes with them. We could have drawn them. A gateway for 'is the appointment within two weeks?', another for 'did the external service have capacity?', a third for 'was the payment taken but not confirmed?' would each have been legal BPMN and would each have made the diagram harder to read without making it more correct, because the rules behind them are not settled in the case study. R07 is the honest example: the case study says urgent clinical concerns must be highlighted immediately

and then admits that the hospital has not agreed a complete set of rules for deciding urgency. Drawing a decision we cannot define would have been worse than recording it as an open point, so both the requirement and the open point sit in the stage specification.

The one assigned requirement is audit and access control. It is listed so that the trace is complete, and it is marked assigned because that is where it belongs: user task assignment, form-level validation and the engine's own audit trail are the mechanisms that satisfy it, and restating them as boxes on a strategy diagram would duplicate controls the diagram cannot express.

Two points about consistency across the models. The SD, with its 28 dependencies, is the wider view: fourteen of its dependency pairs line up with a stage in the BPMN, and the rest describe relationships between external actors that the process does not arbitrate. The SR is the narrower view. It details six of the fifteen actors — the six whose internal reasoning actually drives the design decisions in this release — and the remaining nine are represented at the dependency level, which is where their goals are visible.

3. Acceptance criteria assessment

3.1 First draft

The first set was written from the case study alone, before any of it was tested against the models.

- AC-01 A referral is triggered by an external message and only reaches clinical review once a complete pack is held.
- AC-02 The referral decision, the reason for it and the decision maker are returned to the referring organisation.
- AC-03 A new patient appointment is only arranged after a consultant has accepted the referral.
- AC-04 Clinical, administrative and financial responsibility are distinguishable, and the financial area contains no clinical decisions.
- AC-05 Booking goes through the external scheduling service, the patient is told, and appointments inside two weeks are also telephoned.
- AC-06 Consent is recorded and an unauthorised treatment request never reaches administrative processing.
- AC-07 A treatment appointment is not confirmed before payment or a funding approval is recorded.
- AC-08 Every payment is confirmed by the external provider with a status and a transaction reference.
- AC-09 Clinic letters are processed and dispatched within seven days, and overdue letters enter monitoring.
- AC-10 Enquiries are recorded, classified, prioritised and routed, and call handlers give no clinical advice.
- AC-11 Follow-up appointments are arranged inside the period the clinical team asks for.
- AC-12 Audit records identify the user, date, time and action and cannot be edited by ordinary users.

3.2 What we changed, and why

Each first-draft criterion was then put against the models. Where a criterion could not be judged from the artefact it was meant to describe, we rewrote it rather than leaving it as an aspiration. Two were added once we noticed the first set said nothing about the models themselves.

Ref	Problem with the first draft	Revised criterion
AC-01	It never asks for the request-for-missing-information path, which is the part of the check that has an external effect	The referral entry point exists, and checking the pack and asking the referrer for what is missing are two separately identifiable actions.
AC-05	Two different requirements were bundled together, and the telephone trigger cannot be judged from a strategic diagram	Split. AC-05a: the booking stage exchanges messages with the external scheduling service in both directions. AC-05b: the two-week telephone contact exists as its own message flow, with the trigger recorded in the operational-level specification.
AC-07	It described a data-level condition, which is not something the diagram can show	Restated as a topology check: at the sequence-flow level the funding stage precedes the treatment stage.
AC-09	It demanded a seven-day rule that a strategic model has no way of enforcing	Recorded at the level that can carry it: the letter-processing stage exists in its own right, and the seven-day, one-month and three-month thresholds are stated in the monitoring stage specification.
AC-11	It said nothing about what happens when no follow-up slot fits the clinical window	Added: where no appointment fits the requested period the case is highlighted and passed to the pathway team.
AC-13	New. The first set had no structural criterion at all, so a model could have passed while being unreadable to a tool	The model validates without error: every diagram element resolves to a model element, every connection ends on the boundary of the shape it connects to, each exclusive gateway either has a default flow or a condition on every other outgoing flow, and every flow node sits inside a lane.
AC-14	New. Nothing in the first set required the trace to be complete	The traceability table covers all fifteen actors in the SD and states, for anything not modelled, which level carries it.

3.3 Verification and results

Verification was done by inspecting the model files directly rather than by eye. Three scripts were used: one to parse the BPMN XML and its diagram interchange and test the structural rules, one to walk the sequence-flow graph, and one to test the i* source files for dangling nodes, missing dependency links and overlapping shapes. The results are below.

What was checked	How	Result	Detail
Diagram references resolve to model elements	XML parse	Pass	No unresolved references in the BPMN interchange data
Every flow node has a shape and belongs to a lane	XML parse	Pass	16 of 16 flow nodes accounted for
No sequence flow crosses a pool boundary	XML parse	Pass	All 16 sequence flows stay inside the hospital pool
Every connection ends on the boundary of the shape it connects to	XML parse	Pass after correction	35 of 35 pass. Seven failures were found on the first run: one genuine break (see F1) and six start points left behind by a layout change (see F4)
Exclusive gateways are valid for deployment	XML parse	Pass after correction	Both gateways previously had two outgoing flows with no condition, which the Camunda validator rejects. A default flow and a condition expression were added to each (see F2)
Text annotations are not treated as flow nodes	XML parse	Pass after correction	Two annotations were listed as lane members; annotations are artefacts, not flow nodes (see F3)
No connection passes through a shape	XML parse	Pass	No cases found
Labels do not overlap each other or the connections they belong to	XML parse	One open item	Fifteen message-flow labels currently sit on top of their own connection line and one label sits well away from it (see F5)
Every node is reachable from a start event	Graph walk	Pass	No unreachable or orphaned nodes; all four end events are reachable
IDs are unique and the XML is well formed	XML parse	Pass	No duplicate identifiers; namespaces complete
The SD has no dangling actors or missing links	Model parse	Pass	15 actors, 28 dependencies, 56 dependency links, every actor used
The SR has no dangling nodes	Model parse	Pass	Every node in the SR carries at least one link
Refinement links in the SR mean what they say	Model parse	Pass after correction	All 23 refinements were OR, which reads as 'any one of these will do' for goals where every child is required. Nineteen of them, covering seven goals, were changed to AND (see F6)
No overlapping shapes in the i* models	Model parse	One open item	The SD has two dependency labels overlapping; the SR has one node overlap (see F8)

3.4 Defects found, and what happened to them

The checks above found eight defects. Six were corrected while we were reviewing; two are recorded as open because they are presentation issues rather than behavioural ones. Two further items are listed as scope notes rather than defects, because they follow from the level we chose to model at and are already covered by the traceability table.

Ref	Artefact	What was found	Action
F1	BPMN	The connection from the enquiry start event to the enquiry stage ended about 40 pixels inside the target box instead of on its boundary, so the arrow did not visibly reach the shape.	Corrected. End point moved to the left edge of the target.
F2	BPMN	Both exclusive gateways had two outgoing flows and no condition on either, which Camunda rejects at deployment.	Corrected. Each gateway now has a default flow and a condition expression on the other branch.
F3	BPMN	Two text annotations were listed as lane members. Annotations are artefacts, not flow nodes.	Corrected. The annotations were removed from the lane references.
F4	BPMN	After the external participant pools were repositioned, six message flows still started where the pools used	Corrected. The endpoint check was extended to catch this class of error and all six

		to be.	were realigned.
F5	BPMN	Fifteen message-flow labels sit on top of the line they belong to, and one label is roughly 300 pixels away from its own line. This followed a re-save in the modelling tool, which re-centred the labels.	Open. Label offsets have been recalculated and the correction is ready to apply on the next export. It is a legibility issue only; no connection is affected.
F6	i* SR	All 23 refinements were OR. For seven goals this read as 'any one of these children will do', when in fact every child is needed to satisfy the goal.	Corrected. Nineteen links were changed to AND. The four remaining OR links each have a single child, so they carry no alternative-satisfaction claim.
F7	i* SR	One dependency sends the appointment letter to the patient, while the SD sends it to the correspondence service. The two models disagree about who the hospital depends on.	Open. Reconciling it adds a correspondence actor to the SR, which we have held back to keep the current SR readable; the dependency text is unchanged in the meantime.
F8	i* SD / SR	One dependency label overlap in the SD and one node overlap in the SR ('Report on my treatment' against 'Say yes to treatment').	Open. Both are layout only and both are quick to clear by dragging the affected shapes.
N1	Scope	Refunds, treatment modifications, cycle reviews, cancellations and management reporting are not drawn as separate branches.	Not a defect. They are stated in the specification of the stage that owns them, which is where the operational model will implement them. See R11, R12, R14, R17 and R19.
N2	Scope	Audit and access control do not appear in either model.	Not a defect. They are platform controls, assigned at the level that owns them. See R20.

3.5 Evaluation

The strongest part of this portfolio is the structural side of the strategic model. The responsibility split the case study keeps returning to — clinical judgement, administrative work and financial work kept apart — is expressed in a way that can be checked without reading a word of documentation, and the ordering constraint that money is settled before a treatment appointment is confirmed is carried by the sequence flows themselves. Seven of the eight defects the verification found were in the model's own mechanics rather than in its content, which suggests the content decisions were the sound ones.

The weakest part is the link down to the operational level. Every stage is a call activity carrying a process id, so the connection is declared, but the operational processes do not exist yet and the chain cannot be walked end to end. Until that model is built, the third of the three things the strategic model is supposed to show — organisational responsibility, external interaction and the link to operational processes — is present in form but not in evidence.

On the acceptance criteria, the revision from twelve to fourteen was not a softening. Six criteria were rewritten because they could not be judged from the artefact they described: a seven-day rule that a strategic model cannot enforce, a data-level condition where only a topology check is possible, and two requirements bundled into one. Two were added because the first set said nothing about the models being structurally sound or the trace being complete.

Three things would move this from a well-formed strategic portfolio to a demonstrable one. Completing the operational model and matching its process ids to the stages here would close the traceability chain. Clearing the layout items in F5 and F8 is mechanical and would remove the only findings still open. Reconciling F7, or stating explicitly that the SR is scoped to the six actors whose reasoning drove this release, would settle the last cross-model inconsistency.

4. Supporting files

What the brief asks for	File	Format	Included
Editable BPMN model	UFCEP6-0-3_Hospital_Patient_Pathway_Strategic_R2.bpmn	.bpmn	Yes
Readable PDF export	Portfolio Diagrams.pdf	.pdf	Yes — one page per model, exported from Word, at a page size that keeps

			the labels legible
Editable i* source files	SD diagram.txt / SR diagram.txt	.txt (piStar JSON)	Yes — open directly in piStar
Written assessment in .docx or PDF	This document	.docx and .pdf	Yes — both formats supplied